

Lab 15: Implement Rabin-Miller Primality Testing

Theory:

The Rabin-Miller algorithm is a probabilistic primality test used to determine if a given number is prime. It is widely used in cryptography due to its efficiency and high probability of correctness when run multiple times.

Complexity Overview

Resource	Complexity	Condition
Time	$O(k * \log^3 n)$	k = number of randomized iterations
Space	$O(1)$	Constant memory requirement

Algorithm

1. Handle base cases (small numbers).
2. Find d such that $n - 1 = 2^s * d$.
3. For k iterations, pick a random base a .
4. Calculate $x = a^d \bmod n$.
5. Square x repeatedly. If it doesn't reach $n-1$, return false.

Pseudocode

```
BEGIN
  IF  $n < 2$  THEN return False
  IF  $n = 2$  or  $n = 3$  THEN return True
   $d \leftarrow n - 1$ 
  WHILE  $d \% 2 = 0$  DO
     $d \leftarrow d / 2$ 
  END WHILE
  FOR  $i \leftarrow 1$  to  $k$  DO
     $a \leftarrow \text{random}(2, n-2)$ 
     $x \leftarrow \text{power}(a, d, n)$ 
    IF  $x = 1$  OR  $x = n-1$  THEN continue
    WHILE  $d \neq n-1$  DO
       $x \leftarrow (x * x) \bmod n$ 
       $d \leftarrow d * 2$ 
      IF  $x = 1$  THEN return False
    END WHILE
    IF  $x \neq n-1$  THEN return False
  END FOR
  return True
END
```

Conclusion:

The program successfully utilized the Rabin-Miller probabilistic sorting model to discern prime compositions. Through modular exponentiation combined with dynamic random factorizations over 5 iterative confidence loops ($k=5$), the algorithm sharply narrowed down the mathematical thresholds defining primality across a wide numeric integer scope.

Remarkably, the test accurately tagged natural primes components (2, 3, 5, 17, 1009) as 'Probably Prime'. Concurrently, the logic dynamically triumphed over algorithmic 'Fermat's Pseudoprimes' known as Carmichael numbers (561 and 1729), proving its highly aggressive algorithmic robustness by filtering and correctly marking them as 'Composite'.